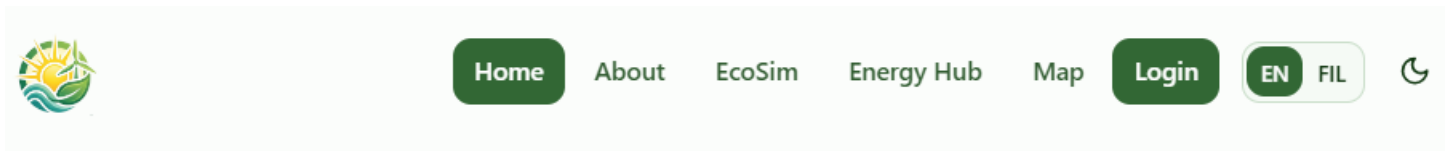


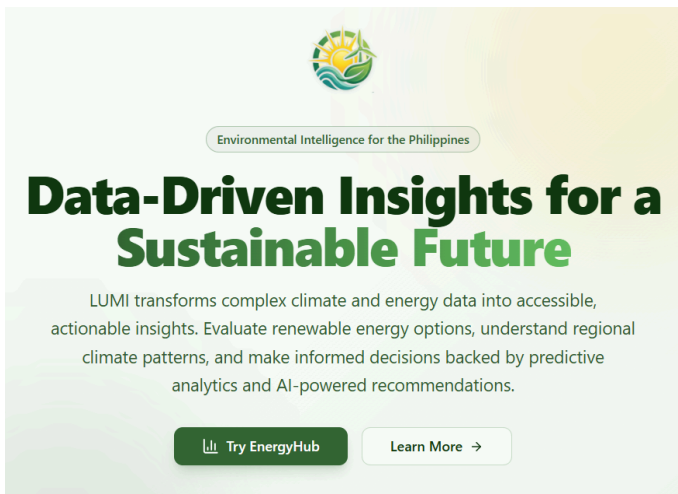
Home

General revision:

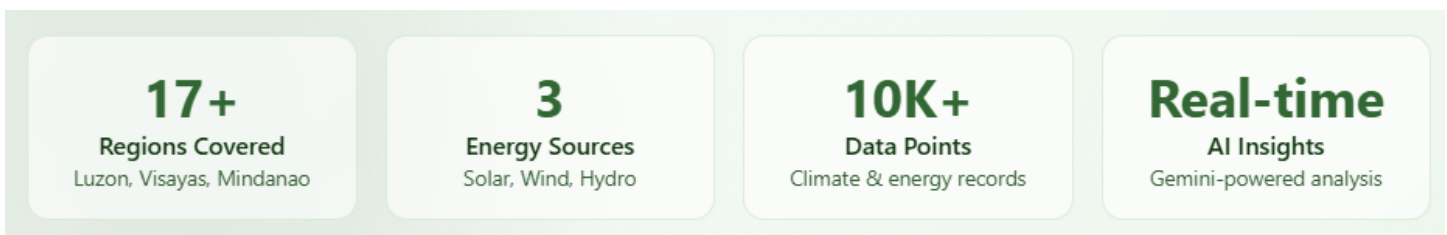
- Mas malaki onti na font and mas dark na font color



1. **EcoSim** → Compare Options (pwede ilagay sa ilalim yung dapat na name for that page)
2. **Energy Hub** → Explore Energy
3. **Login** → Sign in



1. **Data-Driven Insights for a Sustainable Future** → Find the Best Renewable Energy Option for your Area
2. **Description below header** → LUMI helps you compare solar, wind, and hydro energy using weather and energy data from different parts of the Philippines. See which options may work in your area, their estimated costs, and their possible benefits.
3. **Try EnergyHub** → Explore Energy in my Area
4. **Learn more** → How LUMI Works



1. **17+ Regions Covered container** → header: Philippine Regions Covered, description: Explore climate and energy information across the country
2. **3 Energy Sources container** → header: Solar, Wind & Hydro, description: Compare three renewable energy options
3. **10K+ Data Points** → header: Reliable Data Sources, description: Information from recognized weather and energy sources

4. **Real-time AI Insights** → header: Easy-to-Understand Results, description: Get simple explanations and recommendations

PLATFORM MODULES

Intelligent Tools for Energy Decisions

LUMI combines interactive dashboards, predictive forecasting, and AI-driven recommendations to bridge the gap between technical climate data and public understanding.



EnergyHub

Interactive climate and energy dashboard displaying temperature trends, rainfall patterns, energy consumption, and electricity demand across Philippine regions with AI-generated interpretations.

Regional Filtering

KPIs & Charts

AI Insights

NGCP & OpenWeather



Ecosim

A rule-based recommendation and simulation engine that evaluates solar, wind, and hydro feasibility per region. Input your monthly bill and consumption to see estimated costs, savings, and environmental impact.

What-If Scenarios

Cost Analysis

ROI Estimation

Source Comparison



AI Intelligence

Powered by Gemini API, LUMI delivers contextual explanations for every chart, forecast, and recommendation. Natural language insights make complex environmental data understandable for everyone.

Natural Language

Chart Explanations

Recommendations

Comparative Summaries

1. **EnergyHub description** → **Explore energy and weather information in your area.** See temperature, rainfall, electricity use, and other trends through easy-to-read charts and explanations.
2. **EnergyHub labels**
 - **KPIs & Charts** → Charts & Key Information
 - **NGCP & OpenWeather** → remove (add in a different page)
3. **Ecosim description** → **Compare renewable energy options.** Choose a location and compare solar, wind, and hydro based on estimated cost, savings, and local conditions.
4. **Ecosim labels**
 - **ROI Estimation** → Estimated Savings
5. **AI Intelligence** → **Get simple explanations.** LUMI can explain charts, forecasts, and recommendations using everyday language. (Add sa parentheses or what, “Powered by AI”)
6. **AI Intelligence labels**
 - **Natural Language** → Simple Explanations

HOW LUMI WORKS

From Raw Data to Clear Decisions

LUMI simplifies the journey from scattered climate and energy information to confident renewable energy choices.



STEP 1

Data Collection

Aggregates public climate and energy datasets from NGCP, OpenWeather, and regional statistics into a unified, queryable foundation.



STEP 2

Climate & Energy Analysis

Visualizes temperature trends, rainfall patterns, energy consumption, and demand across Luzon, Visayas, and Mindanao.



STEP 3

AI Processing

Applies rule-based logic and Gemini-powered AI to interpret patterns, generate forecasts, and produce contextual recommendations.



STEP 4

Decision Support

Presents actionable insights, what-if simulations, and costed recommendations so users can choose the best renewable path forward.

1. I think it would be much better to show how siya gagamitin ng users, rather than kung paano gumagana yung backend part ng system hehe
2. **Step 1** - header: Choose your Location, description: Select your province, city, or region
3. **Step 2** - header: Explore Local Conditions, description: See rainfall, sunlight, wind electricity use, and other information
4. **Step 3** - header: Compare your Options, description: Compare solar, wind, and hydro based on your area's conditions
5. **Step 4** - header: See your Recommendation, description: Get an easy-to-understand explanation of which options may suit your area

Solar, Wind & Hydro Analysis

LUMI evaluates the feasibility of three core renewable energy sources tailored to Philippine regional conditions, helping users understand suitability, costs, and environmental benefits.



Solar Energy

Evaluates solar irradiance, peak sun hours, and rooftop potential per region. Research shows Filipinos in Metro Manila demonstrate a strong preference for solar over biomass, wind, and small-scale hydropower, and community-based solar initiatives have provided positive impacts in island communities.

PALANCA-TAN ET AL., 2024;
FRANCO & TAEIHAGH, 2024



Wind Energy

Analyzes wind speed patterns and turbine viability for coastal and elevated areas. Studies in the Philippines confirm that combining multi-criteria decision methods with machine learning yields accurate site selection for wind power, providing policymakers and investors with objective feasibility data.

CERNA ET AL., 2023; ABDULLAH
ET AL., 2025



Hydro Energy

Assesses rainfall trends, river flow data, and micro-hydro opportunities. Research on hybrid renewable systems in Southern Philippines identifies hydropower as a viable component alongside solar and wind, with socio-environmental and techno-economic factors determining site suitability.

TARIFE ET AL., 2023

1. **Solar Energy** - sub-header: Best where there is strong sunlight, description: LUMI examines sunlight and local weather conditions to help estimate whether solar energy may work well in your area.
 - Add "Based on the Philippine and international research" clicking it can reveal the citations
2. **Wind Energy** - sub-header: Best in locations with reliable wind, description: LUMI examines wind conditions to help determine whether wind energy may be suitable for your location.
 - Add "Based on the Philippine and international research" clicking it can reveal the citations
3. **Hydro Energy** - sub-header: Best near suitable water resources, description: LUMI examines rainfall and water-related conditions to help determine the potential for small-scale hydropower.
 - Add "Based on the Philippine and international research" clicking it can reveal the citations

Why the Philippines Needs Renewable Intelligence

The Philippine grid relies on fossil fuels for approximately 70.5% of energy production (Gonocruz et al., 2024). In developing countries like the Philippines, financial limitations, governance challenges, and societal resistance remain significant barriers (Zhindon-Almeida & Ruiz-Carrillo, 2025). Public perception and education also significantly shape adoption intentions (Esiri et al., 2024; Aguilera et al., 2024). LUMI closes this gap by transforming fragmented technical data into accessible, cost-inclusive insights.

Read the
Research



1. **Change content** → Much of the country's electricity still comes from fossil fuels. Renewable sources such as solar, wind, and hydro can provide cleaner alternatives, but choosing the right option can be difficult.

LUMI brings weather, energy, and cost information together so people can compare their options more easily.

2. **Read the research** → See Our Data Sources



Start Analyzing Your Energy Future

Whether you are a student, homeowner, policymaker, or researcher, LUMI gives you the clarity to evaluate renewable energy with confidence backed by real data.

⚡ Try Ecosim

📊 Try EnergyHub

1. **Start Analyzing...** → Find the Right Renewable Energy Option for You (description: Choose your location to explore climate information or compare solar, wind, and hydro options.)
2. **Try EcoSim** → Compare Energy Options
3. **Try EnergyHub** → Explore Climate Data

REFERENCES

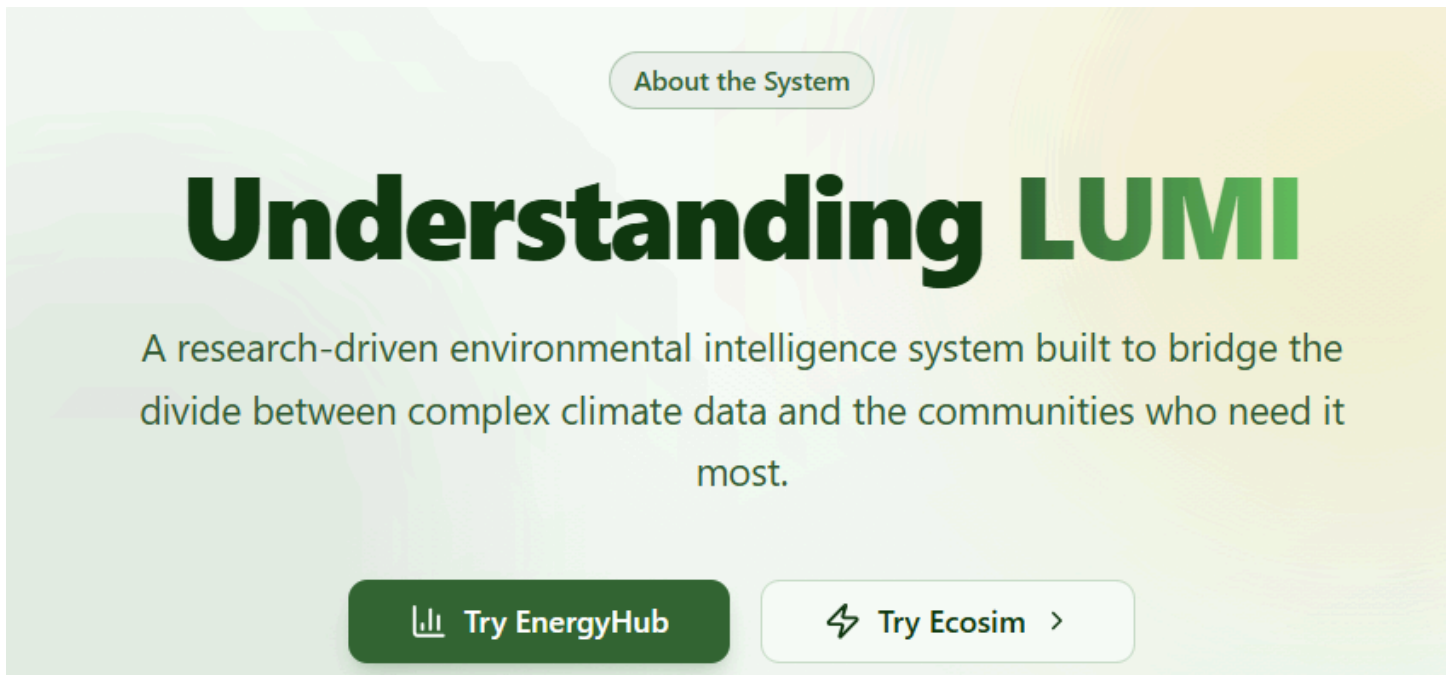
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1. **Change References** → Our Data & Research
2. **Under Our Data & Research** → LUMI uses published research and trusted climate and energy data sources.
3. **Under the LUMI uses...** → button: View Sources & References

About

Feedback:

1. Wag natin siguro ilagay yung mismong citation sa text, pwede siguro na may parentheses after every information na may laman na hyperlink ("View Sources")
2. Same revision with references hehe



1. **Understanding LUMI** → Making Renewable Energy Information Easier to Understand, description: LUMI is a Philippine-based platform that helps people understand climate and energy information, compare renewable energy options, and make more informed decisions.
2. **Try EnergyHub** → Explore Energy Data
3. **Try EcoSim** → Compare Energy Options

Why LUMI Was Developed

The Philippine electrical grid has a capacity of approximately 27.0 GW and is heavily dependent on fossil fuels, which make up around 70.5% of the country's energy production, with coal alone accounting for 43.1%, while only 29.5% comes from renewables (Gonocruz et al., 2024). In developing countries such as the Philippines, financial limitations, governance challenges, and societal resistance have been identified as persistent barriers to renewable energy adoption (Zhindon-Almeida & Ruiz-Carrillo, 2025; Rana et al., 2025). Despite growing awareness, renewable energy remains poorly understood and is often perceived as an expensive investment (Wong et al., 2023).

Furthermore, fragmented climate and energy-related data are often presented in complex formats, hindering public understanding and utilization (Lenain, 2026). Public perception and education significantly shape adoption intentions, and increasing climate awareness can shift sentiment in favor of renewables (Esiri et al., 2024; Aguilera et al., 2024). LUMI was developed to close this gap by transforming overwhelming technical datasets into clear, visual, and actionable insights with cost-inclusive transparency.

Unlike generic information websites, LUMI is an interactive decision-support platform grounded in research on multi-criteria decision analysis and intelligent decision support systems (Beriro et al., 2022; Bączkiewicz et al., 2024), allowing users to simulate scenarios, compare energy sources, and understand regional climate patterns in the context of real energy demand.

70.5%

Fossil Fuel Share

Of the Philippines' energy production comes from fossil fuels, with coal at 43.1% (Gonocruz et al., 2024)

29.5%

Renewable Share

Only about 29.5% of the country's energy is generated from renewable sources (Gonocruz et al., 2024)

3 Barriers

To Adoption

Financial limitations, governance challenges, and societal resistance hinder renewable transition in developing countries (Zhindon-Almeida & Ruiz-Carrillo, 2025)

1. **Why LUMI Was Developed** → Why LUMI Was Created, description: Choosing renewable energy can be difficult. Information about costs, weather conditions, energy use, and available technologies is often spread across different sources and can be hard to understand. LUMI brings this information together in one place and explains it in a simpler way.
2. **Below it are the 3 containers**
3. **Below the 3 containers** → Want to see the evidence? (button: View Research & Sources)

Mission & Vision

LUMI is guided by a commitment to environmental literacy and democratized access to energy intelligence.



Mission

To democratize environmental intelligence by transforming complex climate and energy data into accessible, actionable insights that empower every Filipino to understand, evaluate, and transition toward renewable energy solutions with confidence and transparency.



Vision

A future where data-driven sustainability is within reach of every community. We envision a Philippines where students, households, and policymakers alike make informed energy decisions rooted in clear analytics, reducing fossil fuel dependence and accelerating national climate resilience.

1. **Mission** → To make climate and renewable energy information easier for Filipinos to understand and use when making energy decisions.
2. **Vision** → A Philippines where students, households, communities, and decision-makers can easily access reliable information and make better choices about renewable energy.

The Need for Environmental Intelligence

LUMI is built on a recognized research gap: the absence of user-friendly, data-driven tools for renewable energy evaluation in the Philippines.



Educational Resource

Research shows that education significantly affects people's perception of energy sources and their willingness to transition (Aguilera et al., 2024). LUMI serves as a living case study that makes abstract climate concepts tangible through interactive visualizations, directly supporting classroom and community learning.



Decision Support

Studies confirm that multi-criteria decision analysis improves social impact and policy comprehension in renewable energy planning (Estévez et al., 2021; Witt & Klumpp, 2021). Government agencies can use LUMI to identify high-potential zones and craft strategies backed by structured evaluation rather than assumptions.



Research Groundwork

Environmental intelligence, defined as the use of machine learning and AI to simulate predicted environment models (Bassetti, 2024), is a growing field. LUMI demonstrates the practical application of rule-based AI and predictive analytics in tackling real-world environmental challenges, creating a foundation for future systems.

1. **Change The need for environmental intelligence** → Who can use LUMI?
2. **Change containers** to → significance of the study (beneficiaries)
3. Remove research problem statement

How LUMI Is Structured

LUMI integrates four interconnected modules that guide users from raw environmental data to confident renewable energy decisions.



EnergyHub

Interactive dashboard for climate patterns, energy consumption, and demand trends. Dashboards that use graphical visualizations improve comprehension of complex data ("Dashboard," 2026; What Is Data Visualization?, n.d.), while meteorological data is a critical input for predicting renewable energy output (Das et al., 2022; Bandara et al., 2026).



Ecosim

Rule-based recommendation and what-if simulation engine comparing solar, wind, and hydro with real costing and savings estimates. Studies confirm that technical criteria hold the most weight (53.6%) in renewable energy decision-making, followed by environmental impact (29.0%) (Shatnawi et al., 2021).



AI Analysis

AI-powered natural language explanations for charts, forecasts, and recommendations. Explainable artificial intelligence ensures insights are tailored to user needs and unambiguous (Panagoulas et al., 2023), while AI can effectively optimize renewable energy by predicting supply and demand (Algburi et al., 2025).



Data Visualization

Unified visual analytics including KPIs, trend charts, comparative graphs, and regional overview panels. Data visualization is defined as the representation of data using graphics to make patterns accessible (What Is Data Visualization?, n.d.), and classical forecasting models are sufficient when seasonal patterns are predictable (Mustafa & AI-Yozbak, 2025).

1. How LUMI is Structured
2. **EnergyHub** → Sub-heading: Explore climate and energy data, description: View easy-to-read charts showing weather, electricity use, and energy trends across different Philippine regions.
3. **EcoSim** → sub-heading: Compare renewable energy options, description: Compare solar, wind, and hydro using local conditions, estimated costs, and possible savings.
4. **AI Explanations** → sub-heading: Understand your results, description: Get simple explanations of charts, forecasts, and recommendations.
5. **Data & Charts** → sub-heading: See information clearly, description: Turn complex information into maps, graphs, and comparisons that are easier to understand.

BUILT WITH MODERN TOOLS

Technology Stack

LUMI uses a modern, scalable architecture designed for performance, accessibility, and seamless integration of AI and environmental data services.



Frontend

- React 18 + Vite
- Tailwind CSS
- shadcn/ui Components
- React Router
- Lucide Icons



Backend

- FastAPI (Python)
- RESTful API Design
- Pydantic Validation
- Async Data Processing



Database

- Supabase PostgreSQL
- Row Level Security
- Real-time Sync
- Secure Auth



AI & ML

- Gemini API (Google)
- Rule-Based Engine
- Predictive Analytics
- Trend Forecasting

1. **Add a small section?** → Interested in how LUMI was built? (button: View Technical Architecture)

Designed for Real-World Impact

Every technology choice in LUMI serves the goal of making environmental intelligence fast, accessible, and understandable. From responsive mobile layouts to AI-generated explanations, the architecture supports users across all backgrounds and devices.

Responsive Design

Dark Mode Support

Accessibility First

Performance Optimized

1. **Designed for Real-World Impact** → Made for Everyone
 - Sub-heading: Works on Phones & Computers - description: Use LUMI on different screen sizes

- Sub-heading: English & Filipino - description: Choose the language you're more comfortable with.
- Sub-heading: Easy-to-Read Design - description: Clear text, icons, and explanations.
- Sub-heading: Light & Dark Modes - description: Choose the display that's comfortable for you.

Ready to Explore LUMI?

Dive into regional climate data, run renewable energy simulations, and discover actionable insights tailored to the Philippines.

 **Try EnergyHub**

 **Launch Ecosim**

1. **Ready to Explore?** - description: Check energy and weather information for your area or compare solar, wind, and hydro options.
2. **Try EnergyHub** → Explore my Area
3. **Launch EcoSim** → Compare Energy Options

EcoSim

Renewable Energy Simulation

Ecosim evaluates solar, wind, and hydropower options for your location based on consumption patterns and environmental data.

1

2

3

4

Step 1 of 4

Select your city or municipality so we can analyze your local climate and resources.

Search mode

Municipality

Province

Choose municipality for the most accurate local climate data.

Search municipality

Q BACOR CITY

Selected: BACOR CITY

Next →

Your Inputs

Location

BACOR CITY

Consumption

350 kWh

Bill

PHP 4,780

Savings goal

50%

AI analysis

Yes

1. **Renewable Energy Simulation** → Find the Best Renewable Energy Options for your Area
2. **Description** → EcoSim looks at your location, electricity use, and local environmental conditions to help estimate which renewable energy option may fit you best.
3. 1 → 1 Location
4. 2 → Electricity Use
5. 3 → Savings Goal
6. 4 → Review
7. Search mode → Select area type
8. Municipality → City/Municipality
9. Choose municipality for the most accurate... → We use your location to estimate local solar, wind, and water conditions.

Renewable Energy Simulation

Ecosim evaluates solar, wind, and hydropower options for your location based on consumption patterns and environmental data.



⚡ Step 2 of 4

Tell us how much electricity you use each month so we can estimate your savings.

Monthly consumption (kWh) ⓘ

350

Find this on your electric bill. It is usually the biggest number in the Usage section.

Monthly bill (PHP)

4780

The total amount you pay each month, before any subsidies or discounts.

Your effective rate is about PHP 13.66 per kWh. This helps us estimate your savings accurately.

← Back

Next →

Your Inputs

Location

BACOR CITY

Consumption

350 kWh

Bill

PHP 4,780

Savings goal

50%

AI analysis

Yes

1. Monthly consumption → Electricity used this billing period (kWh)
2. Find this... → Look for "Actual Consumption" on your Meralco bill
3. Add a "Where can I find this on my bill?" help link, and add like a screenshot with encircled portion of the Actual Consumption from a Meralco Bill
4. PHP → ₱
5. Your effective rate... → Based on your inputs, you currently pay about **₱13.66 per kWh**. We use this to estimate your possible savings.

Renewable Energy Simulation

Ecosim evaluates solar, wind, and hydropower options for your location based on consumption patterns and environmental data.

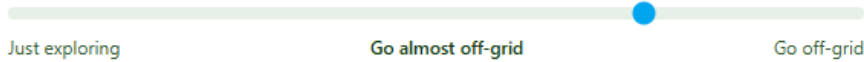


Step 3 of 4

How much of your bill would you like to eliminate with renewable energy?

Desired savings

75%



☒ Include AI analysis

Adds a detailed plain-English explanation of why this renewable source is recommended for your area, along with next steps.

← Back

Next →

Your Inputs

Location

BACCOOR CITY

Consumption

350 kWh

Bill

PHP 4,780

Savings goal

75%

AI analysis

Yes

1. How much of your bill... → How much would you like to reduce your monthly electricity bill?
2. Labels under the slider → Reduce a little → Reduce more → Reduce as much as possible
3. Slider → pwede rin mag-add ng buttons for 25% / 50% / 75% / 100% ?
4. Include AI analysis → Include simple explanation, then add helper text below: Adds a simple explanation of why a renewable option is recommended for your area.

Renewable Energy Simulation

Ecosim evaluates solar, wind, and hydropower options for your location based on consumption patterns and environmental data.



→ Step 4 of 4

Review your inputs and run the analysis.

Location

BACOR CITY

Consumption

350 kWh

Bill

PHP 4,780

Savings goal

75% — Go almost off-grid

We will compare solar, wind, hydro, and geothermal for your location and recommend the best match.

← Back

Running...

Your Inputs

Location

BACOR CITY

Consumption

350 kWh

Bill

PHP 4,780

Savings goal

75%

AI analysis

Yes

1. Either include geothermal sa ibang part ng system or tanggalin dito si geothermal
2. 75% Go almost off-grid → 75% - Reduce most of your electricity cost
3. Option to edit their inputs without having the need to go back

EnergyHub

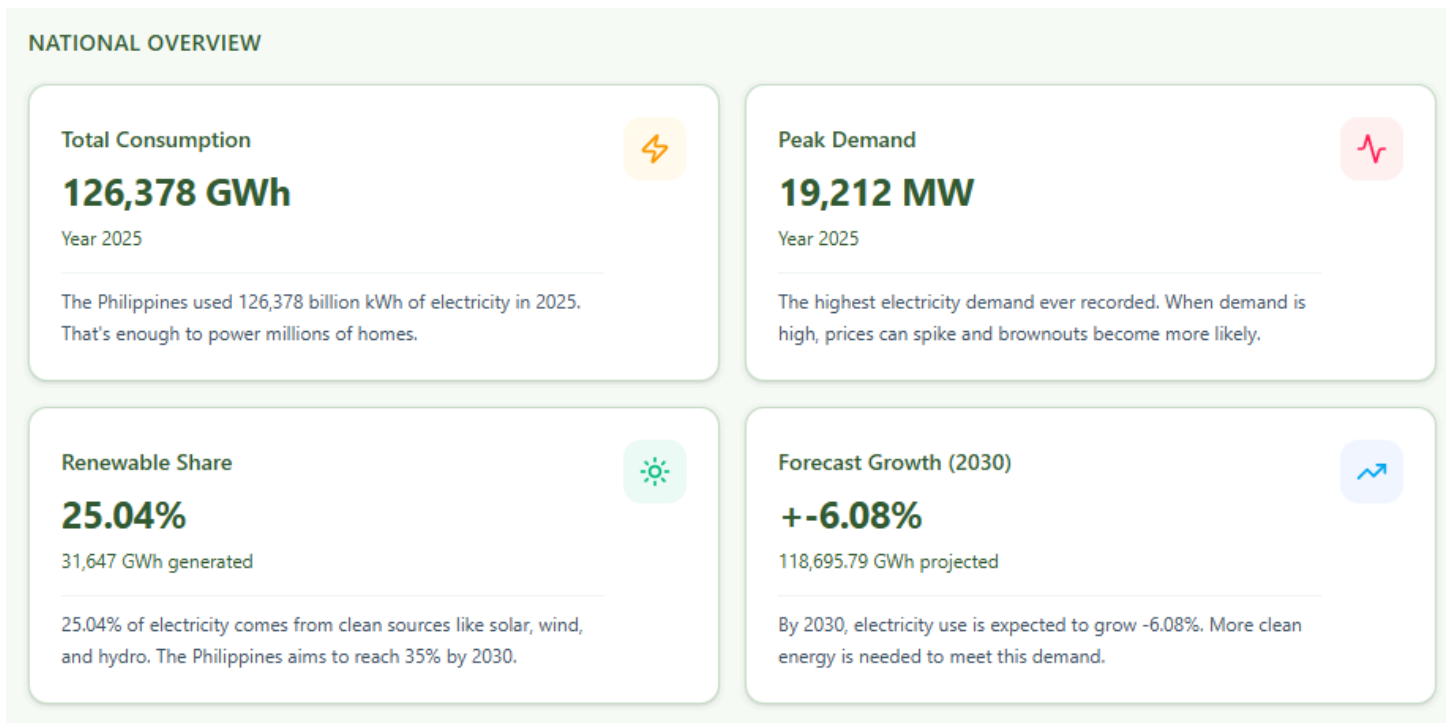
EnergyHub

Explore Philippine national energy statistics, ARIMA-based demand forecasts, and renewable potential across regions. Data sourced from the Department of Energy (DOE) and NASA POWER climate archives.

Disclaimer: This module provides educational insights and is not a substitute for professional energy planning.

1. Explore... → Explore electricity use, renewable energy, and climate information across the Philippines. See past trends, future estimates, and renewable energy potential for different locations.
2. Smaller text below: **Data sources:** Department of Energy (DOE), IRENA, and NASA POWER.

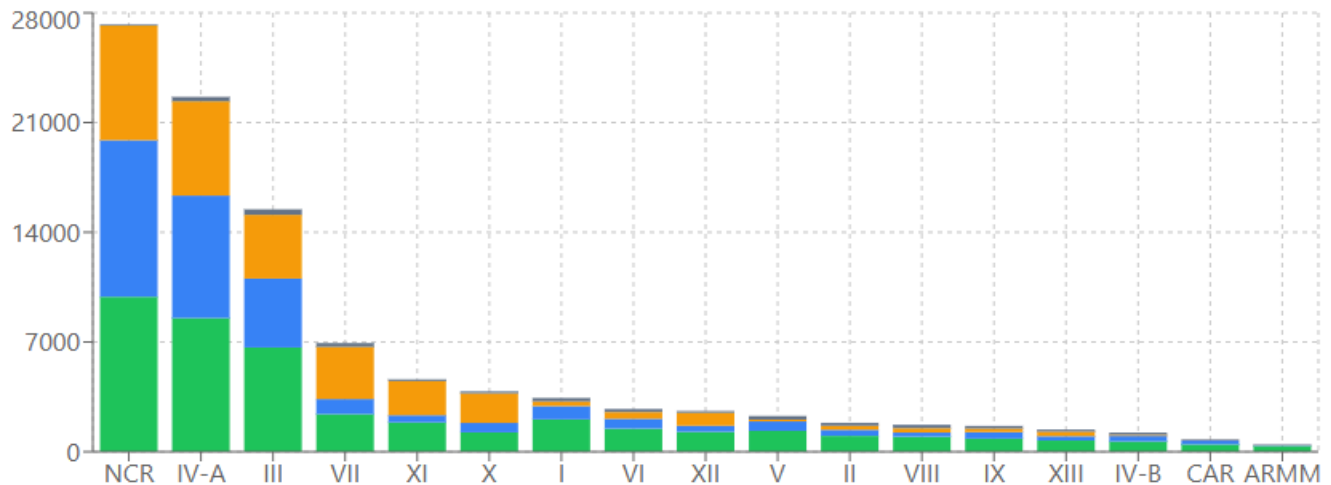
3. Disclaimer... → **For educational use:** EnergyHub helps you understand energy information but should not replace professional advice for energy projects or investments.
4. Quick navigation row? Maybe above or below EnergyHub →
 - Philippines Overview | Explore Map | Energy Trends | Energy Sources | Regional Data



1. Total consumption → Electricity use
2. Small description below → Total electricity used across the Philippines in 2025.
3. Information tooltip → What is GWh?
4. Peak demand → Highest electricity demand
5. Small description below → The highest amount of electricity needed at one time during 2025.
6. Renewable share description → About 1 out of every 4 units of electricity came from renewable sources.
7. Forecast growth description → idk if dapat ba - or +

Provincial Demand (2025)

Electricity consumption by sector per region (GWh). Sourced from DOE Annex 8.



What you are seeing

This chart shows how much electricity each Philippine region used in 2025, broken down by who uses it: homes (green), businesses (blue), factories (orange), and government or street lighting (gray). NCR and Calabarzon (IV-A) tower over the rest because they are the country's economic and population hubs — think of them as the "engine rooms" of the Philippines. Smaller bars do not mean those regions are less important; they simply have fewer people and industries connected to the main grid. Some remote or island regions may also rely more on local generators rather than the national grid, so their numbers here can look lower than their actual energy use.

Why 17 regions?

The Philippines has 17 administrative regions. The data here follows the Department of Energy's 2025 reporting format, which still labels the Bangsamoro region as "ARMM" (its older name). The now-dissolved Negros Island Region (NIR) is excluded because it was re-merged into Western Visayas (VI) and Central Visayas (VII) in 2015.

Values in MWh from DOE Annex 8 (2025).

1. Provincial demand → Electricity use by Region (2025)
2. Add legend above the chart din hehe
3. Add yung region names din pag nag-hover
4. What you are seeing and Why 17 regions? → pwede sigure expandable, para di masyado ma-take up yung space

IRENA Benchmark

Philippines renewable energy statistics from IRENA (cross-reference with DOE data).

Latest RE Capacity

— MW

Latest RE Generation

25,662.58 GWh

RE Share (latest)

20.63%

Data from IRENA. Displayed alongside DOE for benchmarking purposes.

1. IRENA Benchmark → Renewable Energy: International Data
2. Text below: IRENA provides international renewable-energy statistics. Figures may differ from Philippine DOE data because the sources may use different reporting years or definitions.
3. Latest RE Capacity → if unavailable, add “Data currently unavailable”
4. Latest RE Generation → Renewable Energy Generation <data GWh> under it: IRENA | Data year: 0000
5. RE Share (latest) → Renewable Energy Share <data%> under it: IRENA | Data year: 0000
6. Add information tooltip → What is IRENA? → The International Renewable Energy Agency collects renewable-energy statistics from countries around the world.

Energy Choropleth Map

Municipality renewable potential derived from climate & terrain data

Municipality ▼

Solar Potential ▼



Very High (81-100) High (61-80) Moderate (41-60) Low (21-40) Very Low (0-20) No data

1. Energy choropleth map → Renewable Energy Potential by Area
2. Subtitle → See which cities and municipalities have more favorable conditions for solar, wind, or hydro energy.
3. Add label above Municipality → View by:
4. Add label above Solar Potential → Energy source:
5. Add ng location search → instead of manual navigation
6. Add ng explanation or AI insight about the score na meron? (not sure if good idea)

Energy Trends

● Historical ● Forecast

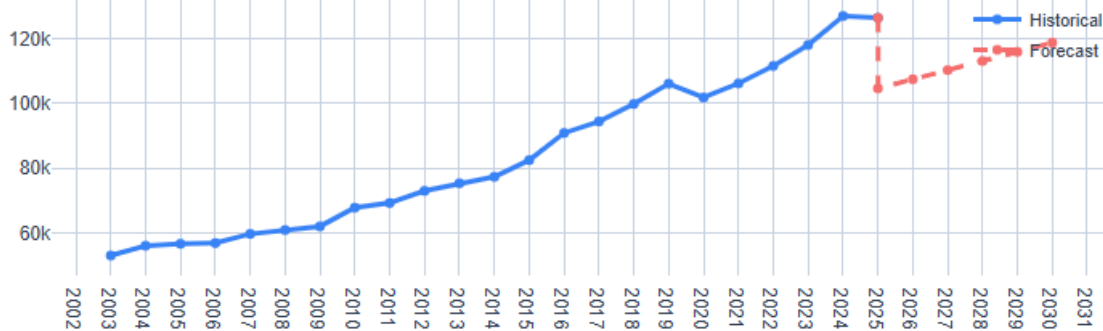
Total Electricity Use Over Time

How much electricity the whole country uses, and where it's heading

What: This chart shows historical national electricity consumption in the Philippines and the forecast through 2030.

Why it matters: Higher consumption means more power plants are needed. It also means more opportunities for clean energy to replace fossil fuels.

What to do: If your region shows above-average growth, consider solar or wind investments ahead of peak-period strain.



AI Explain

1. Yung year, pwede ba pa-vertical siya? Hehe or parang may gap between years → 2002 | 2005 | 2010
2. What: This chart shows... → What you're seeing: How electricity use in the Philippines has changed over time. The solid line shows recorded data, while the dotted line shows estimated future use.
3. Higher consumption... → Changes in electricity use help show whether the country may need more electricity supply in the future.
4. What to do → Possible takeaway
5. AI Explain → Explain this Chart

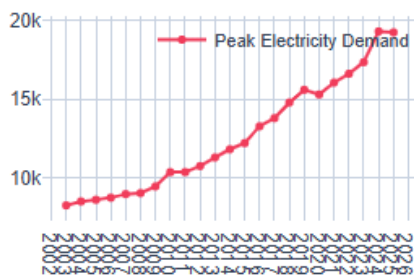
Peak Electricity Demand

The highest amount of electricity needed at any single moment

What: This chart shows the highest recorded electricity demand per year across all Philippine grids.

Why it matters: Rising peak demand means the grid needs more reliable capacity. When peaks are too high, brownouts can happen.

What to do: Municipalities with high peak-demand growth should prioritize distributed solar or battery storage.



AI Explain

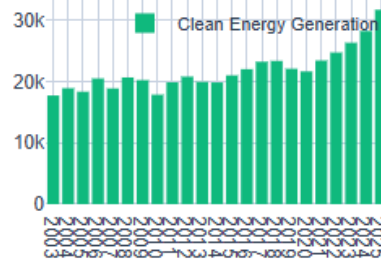
Clean Energy Generation

How much electricity comes from solar, wind, hydro, and geothermal sources

What: This chart shows total renewable energy generation (solar, wind, hydro, geothermal, biomass) per year.

Why it matters: A rising share means the country is successfully replacing coal and gas with clean energy. This helps reduce electricity costs and pollution over time.

What to do: Advocate for local RE adoption if your province lags the national renewable growth trend.



AI Explain

1. What: This chart shows... → What you're seeing:
2. What to do → Possible takeaway
3. Pwede palakihan yung chart?
4. Or gawin na per tab yung sa chart, like may button doon ganun
5. Under ng peak electricity demand → The highest amount of electricity the country needed at one time during each year.
6. Add information tooltip for Peak Demand → Peak demand means the highest level of electricity being used at one time.
7. Clean energy generation → Renewable Electricity Generation
8. Under ng Renewable Electricity generation → Electricity generated from renewable sources such as solar, wind, hydro, geothermal, and biomass.
9. AI Explain → Explain this Chart

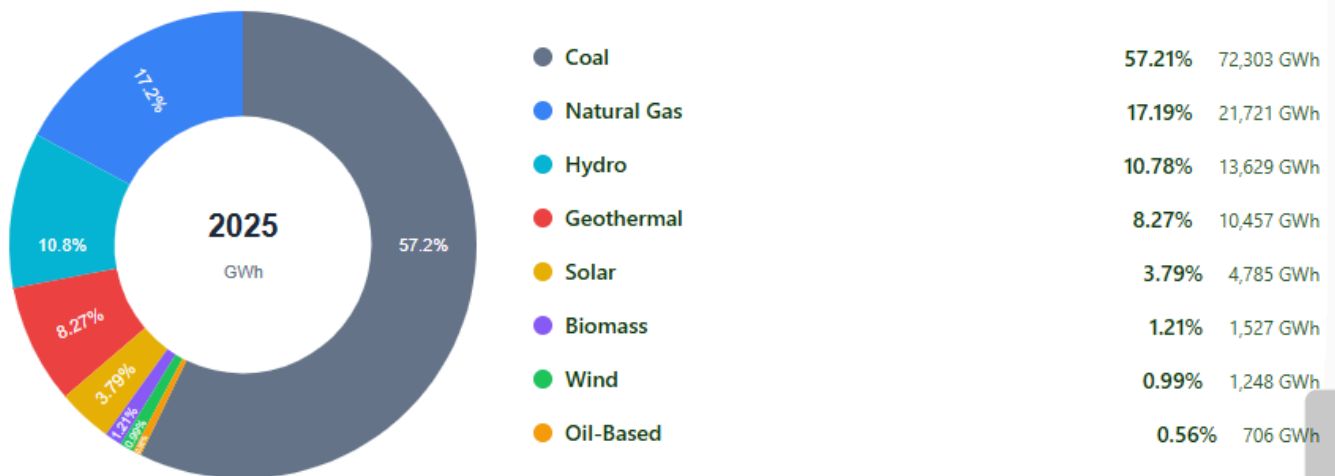
Energy Source Comparison (2025)

Total Generation: 126,376 GWh

What: This donut chart shows what share of total electricity generation comes from each source: coal, natural gas, oil-based, geothermal, hydro, solar, wind, and biomass.

Why it matters: Understanding the mix reveals how dependent the grid is on fossil fuels versus renewables. A high fossil share means higher carbon emissions and exposure to fuel price volatility.

What to do: If fossil sources dominate, push for local solar or wind adoption to reduce household carbon footprint and buffer against fuel price hikes.



1. What: This chart shows... → What you're seeing:
2. What to do → Main takeaway

AI Insight

LLM Mode

Powered by Gemini/Groq LLM — analyzing energy data dynamically.

Overview

Trends

Sources

Map

In 2025, Philippine total electricity consumption reached 126,378 GWh, marking a -0.4% year-over-year change. Renewable energy accounted for 25.0% of the national mix. The capacity margin of 121.6% provides adequate headroom for demand growth in the near term. The ARIMA(1,1,1) model projects consumption will reach 118,696 GWh by 2030, a cumulative growth of -6.1% from 2025.

Continue diversifying the generation mix with cost-effective renewables to maintain long-term supply security.

1. AI Insight → What the Data Means

2. Small text below → Generated with AI based on the energy information shown above.
3. LLM Mode → Detailed Explanation

Map